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Objective:	To convert a measurement from a unit to another unit.
Measurement:	Lengths (Width by length)
1st: Given Unit:	Costumery unit (in. by in.)
To convert to:	Metric unit (cm by cm)
2nd Given unit:	Metric unit (cm by cm)
To Convert to:	Customary unit (in. by in.)
Container used:	Germicidal Wipe Container



First Conversions

Convert 6 inches by 6.75 inches to centimeter by centimeter

Width = 6 inches

Length = 6.75 inches

From Given Tables

$$1 \text{ ft} = 0.3048 \text{ m}$$

$$12 \text{ inches} = 1 \text{ ft}$$

This implies that

$$12 \text{ inches} = 0.3048 \text{ m}$$

$$1 \text{ cm} = 10^{-2} \text{ m}$$

$$1 \text{ cm} = 0.01 \text{ m}$$

We will use Unity Fraction Method to convert the width

We will use the second method to convert the length

Unity fraction method

Convert the Width

6 inches to centimeters

$$6 \text{ in} * \frac{0.3048 \text{ m}}{12 \text{ in}} * \frac{1 \text{ cm}}{0.01 \text{ m}}$$

$$\frac{6 * 0.3048 * 1 \text{ cm}}{12 * 0.01}$$

The answer is 15.24 *cm*

The value on the container is 15.2 *cm*

The calculated value was rounded to the nearest tenth to get the value of the container.

Convert the Length

6.75 inches to centimeters

$$6.75 \text{ in} * \frac{0.3048 \text{ m}}{12 \text{ in}} * \frac{1 \text{ cm}}{0.01 \text{ m}}$$

$$\frac{6.75 * 0.3048 * 1 \text{ cm}}{12 * 0.01}$$

The answer is 17.145 *cm*

The value on the container is 17.1 *cm*

The calculated value was rounded to the nearest tenth to get the value of the container.

Second Conversions

Convert 15.2 centimeter by 17.1 centimeter to inches by inches

Width = 15.2 centimeter

Length = 17.1 centimeter

From Given Tables

$1 \text{ ft} = 0.3048 \text{ m}$

$12 \text{ inches} = 1 \text{ ft}$

This implies that

$12 \text{ inches} = 0.3048 \text{ m}$

$$1 \text{ cm} = 10^{-2} \text{ m}$$

$$1 \text{ cm} = 0.01 \text{ m}$$

We will use Unity Fraction Method to convert the width

We will use the second method to convert the length

Unity fraction method

Convert the Width

15.2 centimeters to inches

$$15.2 \text{ cm} * \frac{0.01 \text{ m}}{1 \text{ cm}} * \frac{12 \text{ in}}{0.3048 \text{ m}}$$

$$\frac{15.2 * 0.01 * 12 \text{ in}}{1 * 0.3048}$$

The answer is 5.984251969 *in*

The value on the container is 6 *in*

The calculated value was rounded to the nearest whole number to get the value of the container.

Convert the Length

17.1 centimeters to inches

$$17.1 \text{ cm} * \frac{0.01 \text{ m}}{1 \text{ cm}} * \frac{12 \text{ in}}{0.3048 \text{ m}}$$

$$\frac{17.1 * 0.01 * 12 \text{ in}}{1 * 0.3048}$$

The answer is 6.732283465 *in*

The value of the container is 6.75 *in*

There is no known applicable rounding done to the calculated value to give the value on the container.

The calculated value was rounded to the nearest whole number to get the measurement on the container.

Reference (MLA Style)

Chukwuemeka, Samuel. "Quantitative Reasoning Project: Measurements and Units on Containers." *Measurements and Units*,
conferencepresentations.appspot.com/Projects/ArithmeticAlgebra/MeasurementsUnits.html. Accessed 25 Sept. 2024.